Supplementary File S6: Integrated Simulator Run

TABLE I
GROUPED CHECK OUTCOMES FROM THE INTEGRATED RUN.

Pipeline stages	Pass	Warn	Fail	Step purpose
P0–P2	11	0	0	Bootstrap, model reference, and provisioning
P3, P4, P4m, P4b–P4f	30	0	0	Acquisition, operation, governance, and records
P6–P7	7	0	0	Seeded objective and multimodal acquisition checks
P8, P8b	16	0	0	Second-scene check and bounded-backstop drill
P9	3	0	0	Platform subsystem checks
P10	9	1	0	BOPET line provisioning and scripted task path
Total	76	1	0	17 phase gates passed

TABLE II
ENDPOINT CHECKS FOR THE IN-RUN `FILM-LINE` CONFIGURATION.

Protocol	Samples	Rejected probes	Legal-write check
Modbus TCP	12	3/3	One accepted
OPC UA	12	3/3	One accepted
MQTT	5	3/3	One accepted
Modbus RTU	11	–	Acquisition only
HTTP	11	–	Acquisition only
Total	51	9/9	Three accepted checks

TABLE III
PROTOCOL-LEVEL MEASUREMENTS RECORDED IN THE INTEGRATED RUN.

Protocol	Samples	Rejected probes	Reported p_{50}/p_{95} (ms)
Modbus TCP	9	6/6	47.673/48.348
OPC UA	11	6/6	21.494/24.175
MQTT	6	6/6	23.015/94.108
HTTP	13	6/6	34.414/47.784
Modbus RTU	15	–	–

This supplement reports the run described in Section V. The accompanying ZIP contains selected machine-readable records, source tables, and figures; it is not a runnable system image.

I. RUN CONDITIONS AND PHASE CHECKS

The run used a deterministic cast-film simulator and tool adapter. No language-model loop or physical equipment was used. The platform and simulator were already running, and the run reused a shared data directory. Its identifier was `20261031034253-1dbo`; the command was `node bench/pipeline.mjs --profile integrated --seed 42 --cl-seeds 3`. The recorded runtime was Node.js 24.19.0 on Windows x64, and the source commit was `60fa983e`. All measurements refer to that snapshot; they do not validate the repository's current HEAD.

All 17 phase gates passed. Of the 77 underlying checks, 76 passed, one raised a warning, and none failed. The warning came from the BOPET semantic-card check in P10; the phase gate itself passed. Table I groups the checks by pipeline stage.

II. ACQUISITION, WRITE PROBES, AND LATENCY

The primary cast-film scene used five simulated endpoints: Modbus TCP, OPC UA, MQTT, HTTP, and Modbus RTU. Five acquisition nodes returned 54 samples. Modbus RTU was an acquisition endpoint without a write-control node. Each of the other four protocol paths received six negative probes, all of which the configured write-admission checks rejected. Each writable path also recorded one accepted legal write. The run did not record all legal-write attempts, so it does not support an estimate of the false-block rate.

The `film-line` configuration was checked in the same run across five endpoints: Modbus TCP, OPC UA, MQTT, Modbus RTU, and HTTP. They returned 12, 12, 5, 11, and 11 samples, respectively (51 total). Each of its three writable paths recorded one accepted legal write and rejected three negative probes (9/9 total). The report recorded no false block or code change. This was an in-run configuration check, not an independent deployment or repetition, and it does not estimate a general false-block rate.

The run stored these latency summaries by protocol. Its log mentions approximately six writes but does not retain the exact per-protocol denominators or event-level timings. The summary's 31.649 ms value is the unweighted mean of the four protocol-level p_{50} values, not a pooled median. These values are descriptive; they do not support adapter rankings or population-level tail estimates.

The same run exercised one approval-enabled write path (P4c). A Line 2 request for configured value 145.5 remained pending until approval returned status 200. The pending write was then released; the report records a simulated PLC readback of 146 and a 452-ms pending-to-approval interval, including polling. No approval-disabled condition was run, so this check documents the enabled path but does not estimate the effect of approval.

III. SEEDED OBJECTIVE AND BOPET TASK

The three fixed-seed cases used the simulator's offline reference $J^* = 89.894$. The simulator score is $J = 55J_{\text{thick}} + 25J_{\text{quality}} + 8J_{\text{energy}} + 7J_{\text{throughput}}$. The thickness term scores deviation from $50 \pm 2\,\mu\text{m}$ with soft degradation outside the band; the quality term reflects defect rate, the energy term screw speed, and the throughput term line speed. The score is defined only for melt temperature in 195–225 °C and pressure no greater than 22 MPa. The runner defines J_{end} as the mean

TABLE IV
FIXED-SEED SIMULATOR OBJECTIVE VALUES. J_0 IS THE INITIAL SCORE; J_{end} IS THE MEAN OF THE LAST TWO VALID TRAJECTORY SCORES, OR THE SOLE VALID SCORE IF THE SEED STOPS BEFORE A SECOND SCORE IS AVAILABLE; J_{last} IS THE FINAL PER-ITERATION SCORE RETAINED IN THE TRAJECTORY.

Seed	J_0	J_{end}	J_{last}	J^*	Ratio	Iter.	Writes
42	66.271	88.775	88.853	89.894	98.8%	2	2
43	86.542	86.542	86.542	89.894	96.3%	0	0
44	63.981	88.384	88.540	89.894	98.3%	2	2

of the last two valid trajectory scores, or as the sole score when the stopping rule ends a case before a second score is available. Table IV also lists J_{last}, the final individual score. Seed 43 began inside the configured stopping region and ended with no iterations or writes. It is a zero-update case and provides no evidence of iterative convergence. Seeds 42 and 44 each used two writes.

The BOPET thickness readings were 28.05, 27.98, 27.98, and $24.84\,\mu\mathrm{m}$, from the initial state through the three writes. The final reading was within the configured $25.0 \pm 0.7\,\mu\mathrm{m}$ band. The task board marked the task COMPLETED, but its associated parameter-change/optimization record remained open and unjudged. This is an integration observation; it does not establish a completed optimization or show that the writes caused the measured change.

IV. PROVENANCE AND INTERPRETATION

The source run did not include a contemporaneous SHA-256 manifest. The ZIP hashes verify the delivered files, not the runtime state at the time of execution.